

THE **AI** CODEBASE ANALYST

FROM ZERO TO A WORKING AGENT

AI Agent Foundations: The Codebase Analyst

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Official Course Plan

Course Philosophy

This is a guided, project-based sprint designed for action. Over 10 focused lessons, you will build, harden, and deploy a single, powerful application from scratch: The AI Codebase Analyst. Each lesson delivers a tangible upgrade to your project, instilling the iterative process of professional AI engineering.

Final Project: The AI Codebase Analyst

You will build and deploy a secure, reliable, and observable AI agent that can:

- Ingest an entire local code repository for context.
 - Answer complex questions about the codebase.
 - Present its findings through an interactive web UI.
 - Be containerized and ready for cloud deployment.
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The 10-Lesson Curriculum

Module 1: The Core Engine (Lessons 1-3)

- **Lesson 1: The Professional Workspace & Your First Observable Call**
 - **Objective:** Establish a professional environment and make a successful, observable LLM call.
 - **Activities:** Set up the project structure, virtual environment, and .env file. Make your first call to an LLM via OpenRouter and view the transaction in LangSmith.
 - **Outcome:** A runnable script that successfully queries an LLM and a link to its corresponding LangSmith trace.

- **Lesson 2: The Agent as a State Machine**

- **Objective:** Build the agent's core logic using the modern LangGraph framework.
- **Activities:** Define the agent's State, Nodes, and Edges. Construct a conversational agent as an explicit state machine.
- **Outcome:** A runnable LangGraph agent that can hold a basic conversation, fully observable in LangSmith.

- **Lesson 3: Giving Your Agent a Simple Tool**

- **Objective:** Enable the agent to act on the world by giving it a tool.
- **Activities:** Use the `.bind_tools()` method to integrate a simple, safe tool.
- **Outcome:** An agent that can answer a question by successfully reasoning about and calling a tool to get the answer.

Module 2: The Knowledge Base (Lessons 4-5)

- **Lesson 4: The RAG Pipeline - Part 1: Ingestion**

- **Objective:** Build the agent's long-term memory from a corpus of external documents.
- **Activities:** Implement the "Load, Split, Embed, Store" stages of a RAG pipeline. Write a standalone script to create a local FAISS vectorstore.
- **Outcome:** A searchable vectorstore containing the embedded knowledge of your target codebase.

- **Lesson 5: The RAG Pipeline - Part 2: Integration**

- **Objective:** Connect the agent's brain to its long-term memory.
- **Activities:** Build a `codebase_retriever` tool that queries the vectorstore. Integrate this powerful new tool into your agent.

- **Outcome:** The agent can now answer complex questions by intelligently retrieving relevant code chunks.

Module 3: The Production Pillars (Lessons 6-8)

● Lesson 6: The Reliable Analyst - Enforcing Structured Output

- **Objective:** Make the agent's output predictable, stable, and machine-readable.
- **Activities:** Use Pydantic models to define a strict output schema and force the LLM to return validated JSON.
- **Outcome:** An agent that produces reliable JSON objects, making it a trustworthy component for any larger system.

● Lesson 7: The Secure Analyst - Building Guardrails

- **Objective:** Harden the agent against common security vulnerabilities like prompt injection.
- **Activities:** Implement a "Constitution" via a system prompt and "sandbox" a tool with path validation. Write an automated test to verify the safeguards.
- **Outcome:** A more secure agent and an automated test proving its resilience against specific attacks.

● Lesson 8: The Resilient Analyst - Handling Failure

- **Objective:** Make the application robust enough to handle real-world errors gracefully.
- **Activities:** Refactor tools with "Lazy Loading." Wrap external API calls with automatic retries and caching.
- **Outcome:** An application that doesn't crash on failure and performs more efficiently.

Module 4: The Final Product (Lessons 9-10)

- **Lesson 9: The Interactive Analyst - Building the UI**

- **Objective:** Create a polished, user-friendly web interface for your agent.
- **Activities:** Use Chainlit to rapidly build a chat application. Connect the UI to the now-hardened agent backend.
- **Outcome:** A professional-looking, interactive web application for your agent.

- **Lesson 10: The Production Pipeline: CI & Deployment**

- **Objective:** Package the entire application for portable, one-command deployment.
- **Activities:** Formalize the security test as a CI check and containerize the complete application with Docker.
- **Outcome:** A complete, containerized application ready for one-command deployment to any cloud provider.